Chapter 3: Distribution Functions

Part I — Statistics Foundation for ML

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Abstract

A distribution is described by a function. This chapter is about those functions: the cumulative distribution function that every random variable has, the mass or density function that most have, and the rules for turning one random variable into another. The payoff is concrete. The *probability integral transform* lets you manufacture a sample from any distribution out of nothing but uniform random numbers — which is how a computer draws from an exponential, a Cauchy, or a logistic — and the *change-of-variables formula* gives the density of any smooth transform of a variable you already understand.

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1 The distribution function and the density

Every random variable, discrete or continuous, is pinned down by one function.

Definition 1.1 (Cumulative distribution function)**.** The **cdf** of X is $F(x) = P(X \le x)$. It is non-decreasing, right-continuous, and satisfies $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$.

A **discrete** variable is described by its **probability mass function** $f(x) = P(X = x)$, and F is a sum of jumps. A **continuous** variable is described by a **probability density function** f with

$$F(x) = \int_{-\infty}^{x} f(t)\,dt, \qquad f(x) = F'(x).$$

The density is not a probability — it can exceed 1 — but integrates to probability over any interval: $P(a \le X \le b) = \int_a^b f$.

Example 1.1 (Exponential)**.** For $X \sim \text{Exponential}(\lambda)$, $F(x) = 1 - e^{-\lambda x}$ and $f(x) = \lambda e^{-\lambda x}$ for $x \ge 0$. Differentiating F returns f, as it must.

The cdf has one property that looks like a curiosity and turns out to be a workhorse: feed a continuous random variable through its own cdf and you get a uniform.

Theorem 1.1 (Probability integral transform)**.** If X is continuous with cdf F, then $U = F(X) \sim \text{Uniform}(0, 1)$. Conversely, if $U \sim \text{Uniform}(0, 1)$ then $X = F^{-1}(U)$ has cdf F.

The converse is how a computer samples from *any* distribution: draw a uniform and push it through the inverse cdf. One stream of uniforms becomes whatever you like, depending only on the F^{-1} you apply:

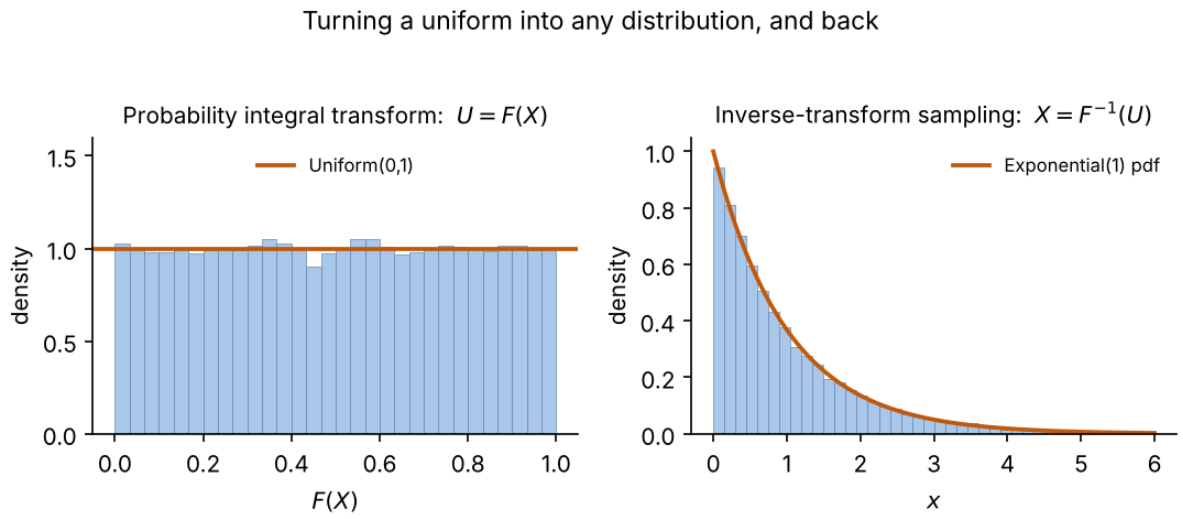
```
import random, math

def inverse_transform(Finv, rng, n):
    return [Finv(rng.random()) for _ in range(n)]        # F^{-1}(U) ~ target

exp_inv      = lambda u: -math.log(1 - u)                # Exponential(1)
cauchy_inv   = lambda u: math.tan(math.pi * (u - 0.5))   # standard Cauchy
logistic_inv = lambda u: math.log(u / (1 - u))           # standard Logistic

rng = random.Random(0)
for name, Finv in [("Exponential", exp_inv), ("Cauchy", cauchy_inv),
                   ("Logistic", logistic_inv)]:
    s = sorted(inverse_transform(Finv, rng, 100000))
    print(f"{name:12s} sample median = {s[len(s)//2]:+.3f}")    # 0.69, 0, 0
```

The same uniform stream, three inverse cdfs, three different distributions. The two directions — data into a uniform, uniforms into data — are shown below for the exponential.



2 Transforming a random variable

When $Y = g(X)$ for a strictly monotone, differentiable g, the density of Y follows from the cdf by one differentiation.

Theorem 2.1 (Change of variables, monotone case)**.** If g is strictly increasing and differentiable with inverse $h = g^{-1}$, then

$$f_Y(y) = f_X\big(h(y)\big)\,\big|h'(y)\big|.$$

Proof. For increasing g, the event $g(X) \le y$ is the event $X \le h(y)$, so

$$F_Y(y) = P\big(g(X) \le y\big) = P\big(X \le h(y)\big) = F_X\big(h(y)\big).$$

Differentiating with the chain rule gives $f_Y(y) = f_X(h(y))\,h'(y)$. For a decreasing g the inequality flips and the same calculation produces $-h'(y)$; the absolute value covers both cases. □

The probability integral transform is now a one-line corollary, taking $g = F$ itself.

Theorem 2.2 (Proof of the probability integral transform)**.** For continuous X with cdf F and any $u \in (0,1)$,

$$P\big(F(X) \le u\big) = P\big(X \le F^{-1}(u)\big) = F\big(F^{-1}(u)\big) = u,$$

which is the cdf of Uniform$(0,1)$. Hence $F(X) \sim$ Uniform$(0,1)$.

3 Non-monotone transforms and higher dimensions

When g is not one-to-one, several x values map to the same y, and the density of Y collects a contribution from each. If $g(x) = y$ has solutions $x_1, \dots, x_k$,

$$f_Y(y) = \sum_{j=1}^{k} \frac{f_X(x_j)}{|g'(x_j)|}.$$

Example 3.1 (Square of a normal is chi-square)**.** Let $Z \sim N(0,1)$ and $Y = Z^2$. For $y > 0$ the equation $z^2 = y$ has the two roots $z = \pm\sqrt{y}$, each with $|g'(z)| = 2\sqrt{y}$, so

$$f_Y(y) = \frac{\phi(\sqrt{y}) + \phi(-\sqrt{y})}{2\sqrt{y}} = \frac{1}{\sqrt{2\pi y}}\, e^{-y/2},$$

using $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$ and its symmetry. This is the χ^2_1 density — the reason squared normal errors turn into chi-square statistics throughout the course.

In higher dimensions the derivative becomes a Jacobian determinant: for an invertible smooth map $Y = g(X)$ with inverse h,

$$f_Y(y) = f_X\big(h(y)\big)\,\big|\det J_h(y)\big|,$$

where J_h is the matrix of partial derivatives of h. The same idea drives the normalizing-flow models of Part III, which chain many invertible maps and track the log-determinant at each step.

A last subtlety makes inverse-transform sampling completely general. When F has flat stretches or jumps — a discrete or mixed variable — it has no ordinary inverse, so one uses the **quantile function** $F^{-1}(u) = \inf\{x : F(x) \ge u\}$. With this generalized inverse the probability integral transform still holds, which is why the same draw-a-uniform recipe samples discrete distributions too. Separating the uniform "dependence" from the marginal F^{-1} in several dimensions is exactly the idea behind **copulas**.

Summary

Every random variable has a cdf F; continuous ones also have a density $f = F'$. Pushing a continuous variable through its own cdf yields a uniform (the probability integral transform), and pushing a uniform through F^{-1} yields the variable back — the basis of random sampling. A smooth transform $Y = g(X)$ has density $f_X(h(y))|h'(y)|$ in one dimension and $f_X(h(y))|\det J_h(y)|$ in several, summed over branches when g is not one-to-one.